



Interleukin-4 receptor signaling and its binding mechanism: A therapeutic insight from inhibitors tool box



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ABSTRACT

Studies on Interleukin-4 (IL-4) disclosed great deal of information about its various physiological and pathological roles. All these roles depend upon its interaction and signaling through either type-I (IL-4R α /common γ -chain) or type-II (IL-4R α /IL-13R α) receptors. Another cytokine, IL-13, shares some of the functions of IL-4, because both cytokines use a common receptor subunit, IL-4R α . Here in this review, we discuss the structural details of IL-4 and IL-4R α subunit and the structural similarities between IL-4 and IL-13. We also describe detailed chemistry of type-I and type-II receptor complexes and their signaling pathways. Furthermore, we elaborate the strength of type-II hetero dimer signals in response to IL-4 and IL-13. These cytokines are prime players in pathogenesis of allergic asthma, allergic hypersensitivity, different cancers, and HIV infection. Recent advances in the structural and binding chemistry of these cytokines various types of inhibitors were designed to block the interaction of IL-4 and IL-13 with their receptor, including several IL-4 mutant analogs and IL-4 antagonistic antibodies. Moreover, different targeted immunotoxins, which is a fusion of cytokine protein with a toxin or suicidal gene, are the new class of inhibitors to prevent cancer progression. In addition few small molecular inhibitors such as flavonoids have also been developed which are capable of binding with high affinity to IL-4R α and, therefore, can be very effective in blocking IL-4-mediated responses.

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1. Introduction

IL-4, a cytokine discovered in the 1980's, remains in the spotlight as a target for designing a potent small molecular inhibitor. This cytokine is primarily produced by basophils, mast cells, and eosinophils, beside Th2 lymphocytes [1]. Its molecular weight varies between 12 and 20 kDa because of irregular pattern of post-translational modification, particularly glycosylation. This cytokine shares most of cell surface receptors, sequence homology, intracellular signaling and limited functional effects with the IL-13 cytokine [2]. Practically, IL-4 is involved in determining cell proliferation, expression of various genes, and in preventing apoptosis in various cell types like lymphocytes, macrophages, epithelial cells, fibroblasts, as well as endothelial cells [1,2].

Naïve helper T lymphocytic cell differentiation and transformation to the T-helper-2 cells (Th2) is the function of IL-4 protein along with production of series of other cytokines including IL-13, IL-5 and IL-10 [3]. It vigorously ceases T helper-1 (Th-1) cell differentiation and down regulates their IFN production. The second dominating function of IL-4 is to regulate immunoglobulin class switching, especially the expression of IgE and IgG4 in human B-cells [4], and IgG1 and IgE in mouse B-cells [5]. IgE production is reduced to 100-fold or more in IL-4, IL-4 receptor (IL-4R) and STAT-6 (substrate of IL-4Rα) [6] knockout mice. While STAT-6 knockout mice infected with helminthic parasites are devoid of differentiating naïve T-cells to IL-4 producing Th-2 cells [6e]. These physiological functions made IL-4 an ultimate player in regulation of allergic conditions. It also acts as a primary source in providing protective immune response against extra-cellular parasites and helminthes. Moreover, in various clinical and experimental trials, it also appears to be responsible for tissue damaging auto immunity effects [7].

IL-4 has a diverse role in hematopoietic tissues by serving as a co-mitogen for B cell growth [8]. It increases the expression of IL-4 receptor [9], CD 23 [10], and class II MHC molecules [11] along with enhanced ThY-1 expression on B cells in combination with lipopolysaccharides [12]. It can prolong the life of B and T lymphocytes in culture without acting as growth factor by itself for resting lymphocytes [13], and can avert apoptosis on IL-4 expressing cells by factor dependent myeloid lineages [14].

IL-4 also has a significant role in tissue adhesion and inflammation. It causes vascular endothelial cells to express vascular cell adhesion molecule-1 (VCAM-1) with tumor necrosis factor (TNF) [15], and reduces expression of the E-selectin protein molecules [16]. This alteration in correspondence to the expression of adhesion molecules by IL-4 is assumed to mediate T-cells and eosinophil recruitment to the site of inflammation rather than other granulocytes [17].

Aberrant expression of IL-4 in number of pathological conditions, underscore its role not only in normal physiology but also in pathogenesis of number of diseases. Therefore, this review will focus on the therapeutic aspects of IL-4 and its receptor, where we will discuss its importance in various disorders such as allergic asthma, different cancers and HIV infections. We will also elaborate various type of inhibitors designed against IL4 and its receptor for treatment of the above mentioned diseases.

2. Biological aspects of IL-4

a) IL-4 Synthesis

In normal hematopoietic cells, IL-4 is synthesized by two different types of cells. First one are the matured lymphoid cells that synthesize IL-4 through specific antigen stimulation, and in turn the soluble IL-4 enhances the synthesis of IL-4 by Th2 cells [3a]. While second type of cells are the mast cells from myeloid cell lineage that developed and differentiated by different cytokines

within the bone marrow and fully matured in their resident tissues. These mast cells do not require prior antigen exposure, but rather secrete pre-formed IL-4 hoarded in granules after binding of high affinity IgE with their cell surface receptors, and can also trigger new gene transcription for IL-4 synthesis [18].

b) IL-4 Gene transcription

IL-4 gene promoter region, which consists of 300 base pairs, harbors Nuclear Factor of Activated T-cells (NF-AT) transcription factor binding region [19]. Likewise, there are minimum five distinct members within the NF-AT family. These transcription factors work in a sequential order similar to NFκB, and remain as dormant cytoplasmic factor that go through calcineurin-dependent dephosphorylation after cell activation. This activation leads to its translocation to the nucleus where it is integrated with AP-1 transcription factor and ultimately results in initiation of IL-4 gene transcription [20].

IL-4 synthesis is focused around three transcription factors: STAT-6, GATA-3 and C-maf. First of all, Th2 cells must receive activation signals to provoke more IL-4 synthesis. The STAT-6 is a fundamental transcription factor for IL-4 receptor signaling and is involved in enhancing IL-4 synthesis. Therefore, the STAT-6 knockout mice were found deficient of yielding Th2 cells [6d]. GATA-3, just like STAT-6, is crucial for Th2 development and can interact with the distal region of IL-4 promoter gene [21]. Lastly, C-maf, a remote relative of AP-1 family, is particularly expressed by Th2 cells rather than Th1 cells [22]. Cells with C-maf deficiency have extremely reduced IL-4 expression, retaining normal expression of other Th2 cytokine. C-maf interacts with the proximal part of IL-4 promoter gene, and any manipulations in this region annihilate IL-4 synthesis completely [18].

3. IL-4 receptor complex scheme

Type I receptor complex was discovered first as a complex between IL-4Rα and γ chain [23]. Most of the hematopoietic, epithelial, endothelial, fibroblast, muscle, hepatocytes and brain tissues cells express IL-4 receptors at the rate of 100–5000 per cell which include [24a,24b]. Majority of cells display type I or type II receptors while a few cells express both types of the receptors.

In the type-I receptor complex, IL-4 first binds with the 140 kDa IL-4Rα chain with high affinity (Kd = 20–300 pM). Physiologically, the signaling cascade starts after hetero-dimerization of IL-4/IL-4Rα complex with the third γ chain (Fig. 1 (left)), while artificial homo-dimerization of IL-4Rα chain produces biochemical signaling within cells [25]. The second chain of IL-4 receptor i.e. common γ chain, which was initially discovered as an integral part of IL-2

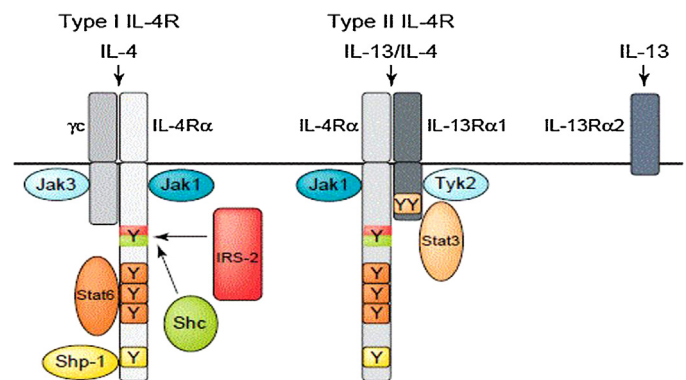


Fig. 1. Represents two type of IL-4 receptors. Left panel explaining type-I receptor with γ-chain while right panel representing type-II receptor with IL-13Rα shared with IL-4 and IL-13 cytokines. (Chatila, T.A., Interleukin-4 receptor signaling pathways in asthma pathogenesis. *Trends in molecular medicine* 2004, 10 (10), 493–499).

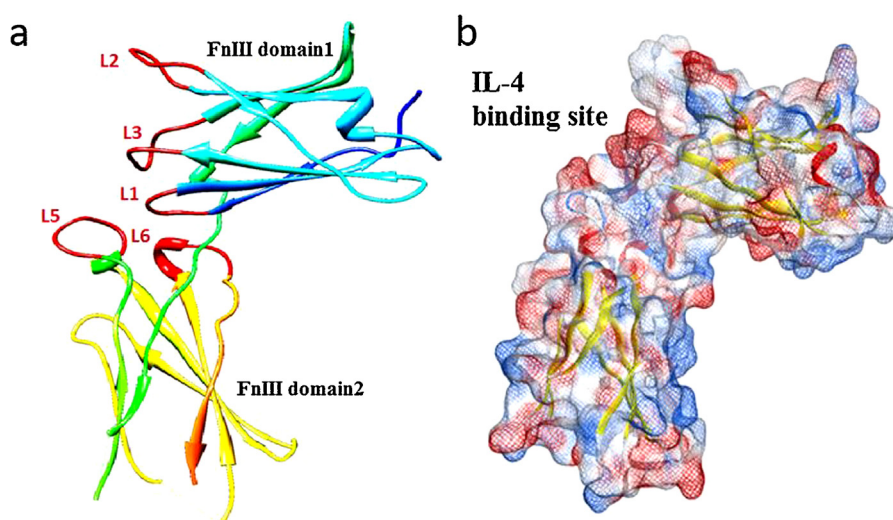


Fig. 2. (a) Ribbon diagram of IL-4 endogenous cytokine protein, consists of four α -helices and three loops. (b) A surface diagram displaying electrostatic potential of IL-4 measured by MOE, representing an intense positive potential for IL-4R α at AC face.

receptor, seems to be preminent chain associated with heterodimerization in various cell types [23,26]. Molecular binding investigation revealed that the γ chain recognizes IL-4/IL-4R α complex [27]. The activation of signaling cascade requires γ chains to moderately enhance the recognized affinity of IL-4R α for IL-4 after IL-4 binding [23a].

In the type-II receptor complex, this γ chain is altered by low affinity IL-13R α -1 (Fig. 1 (right)) which associates with IL-4R α chain [28]. IL-4R α chain also serves as a fundamental part of IL-13 receptor [29]. This receptor exploits other polypeptides of the cell surface rather than a common γ chain [29b,29d,30]. A number of cell lines with defective common γ chain are apparently IL-4 susceptible [29c,31]. IL-13R α expression in these cell lines promoted the probability that IL-13R α may serve as component of IL-4R α complex in non-hematopoietic cells [29d]. IL-4R α is not capable of binding with IL-13, but its interaction with IL-13R α -1 triggers a signaling cascade and makes cells responsive to both IL-4 as well as IL-13. This reason accounts for the fact that these two cytokines share several biological activities [32].

3.1. 3D-Structure of IL-4

IL-4 is a cytokine of four-helix complex [33], represented by inter-connecting anti-parallel helix (A–D) and two elongated intertwined loops i.e. AB and CD which are linked by a little h-sheet wrapped across B and D helices as displayed in Fig. 2a [34]. Four distinct exons are responsible for transcription and execution of AB loop, helix A, helix BC hair pin, helix D and loop CD. The helix BC hairpin appears to be preserved by a parallel disulphide bond between C46 and C49. IL-4 acquires two binding epitopes; one side comprises of A and C helices which binds with high affinity to IL-4R α [34f,35], and the other site involves A and D helices that bind with low affinity to both γ chain and IL-13R α -1 chain [27,35,36].

In various precedents, the biological activity of IL-4 is limited because of the residues diversity among various species. Human IL-4R α does not interact with the IL-4 of murine origin [37]. While it is noteworthy here that the vital binding elements i.e. E9, R53, Y56, R88 remain conserved among species. Side chains that do not confer dominantly in binding, modify the binding specificity between IL-4 and IL-4R α . Numerous basic residues such as K12, R53, R75, K77, R81, K84, R85 and R88 provide massive positive

charge at the face of AC helix of human IL-4 (Fig. 2b). Conversely, IL-4R α chain carries acidic residues within the binding pocket. This charge complementarity leads to the incorporation of IL-4 and IL-4R α through electrostatic steering [38], and the association rate constant for this event is remarkably high, i.e., $>10^7/\text{M s}$ [37].

3.2. 3D-Structure of IL-4R alpha chain

The extracellular binding domain consists of a stretch of 100 amino acids which involves two fibronectin type–III (fnIII) territories, linked by a precise linker moiety. The first terminal of fnIII domain (D1) involves conserved disulphide bonds establishing the h-type of an immunoglobulin fold. While the second fnIII (D2) domain presents s-type fold, however, fnIII topology has no disulphide bond in it [39]. The extra-cellular binding domain of IL-4R α resembles the nominal form of a characteristic cytokine-binding homology region (CHR). In extension, D2 domain also carries ws–ws motif, which is classical for cytokine receptor type I family [40].

The elbow part between two domains is engrossed by an elongated inter-beta sheet loops L2 and L5 that are surrounded by L1 and L3 loops in the first domain and L6 in the second fnIII domain. The linker, L4 bridging the two fnIII domain is hidden beneath linking loops (Fig. 3a) and is not exposed to cytokine ligand, and not involved in IL-4/IL-4R α interaction. IL-4R α associates with IL-4 by a number of acidic (negatively charged) residues in the elbow region (Fig. 3b). Different mutational studies demonstrated their relative affinity by complementary charges (app ~ 10 fold) between IL-4 and its receptor [38,41].

3.3. Comparison between receptor complexes

The receptor complex built by association of composite cytokine-receptor binding surfaces display remarkable differences in the extent and dissemination of polar and non-polar surfaces. As described in previous studies, Sherry L. Laporte et al. explored thermodynamics of this complex assembly in order to describe whether this structural variability is reflected in distinct binding chemistries or not by using isothermal titration calorimetry (ITC) with soluble extra cellular domain of the receptor [2f].

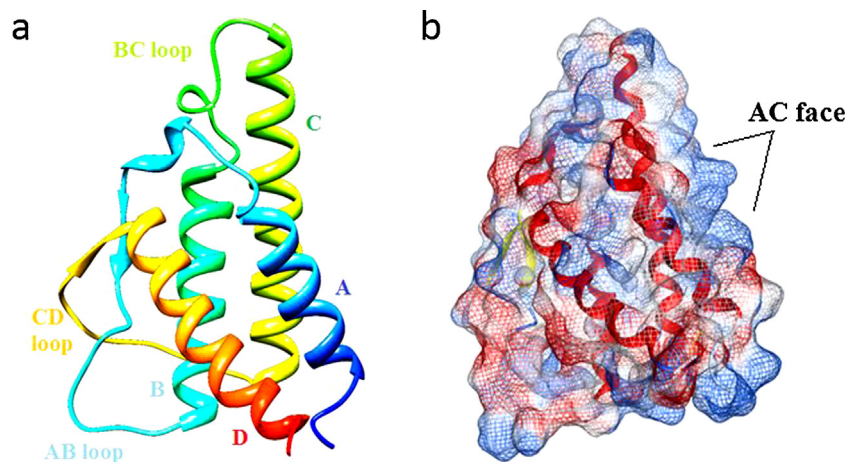


Fig. 3. (a) Ribbon diagram of IL-4R α with two fibronectin domains, each comprising of six β -sheet. The loop region in red color shows areas where IL-4 ligand binds. No coordinates are available for the loop regions from residue 107 to 112 and 163 to 169 in the second FnIII domain due to lack of electron density. (b) Surface diagram of IL-4R α showing electrostatic potential calculated by MOE 2013. Strong negative potential was observed complementary to intense positive potential at IL-4 ligand AC face.

Previous studies [23a,29c,42] deduced that, in type-I receptor complex IL-4 fastens with IL-4R α first and then binds with γ chain consecutively to form a ternary complex (Fig. 4a). Whereas, for type-II receptor complex, IL-4 first joins with IL-4R α and then couples with IL-13R α -1 (Fig. 4b). Inversely, type-II receptor complex with IL-13 results when IL-13 first associates with IL-13R α -1 having affinity of $K_d = 30$ nM, and makes it susceptible for interaction with IL-4R α (Fig. 4c) [42]. The same IL-4 and IL-4R α binary complex incorporate γ chain or IL-13R α -1 remarkably in type-I and type-II receptor complexes by opposite thermodynamics, in which type-I complex exhibits vigorous enthalpy ($\Delta H = -11.7$ kcal/mol) and obnoxious entropy ($\Delta S = -10.5$ kcal/mol) as mentioned in Fig. 4a, while type-II complex in Fig. 4 illustrates poorly preferred enthalpy ($\Delta H = -4.8$ kcal/mol) and robust entropy ($\Delta S = 13.0$ kcal/mol). Moreover, comparative analysis of thermodynamic properties at interfacial sites revealed strong binding affinity of IL-4 with IL-4R α ($\Delta H = -11.2$ kcal/mol, $\Delta S = +3.5$ kcal/mol) versus weak affinity of IL-13 $\Delta H = -5.8$ kcal/mol, $\Delta S = +16.0$ kcal/mol (Fig. 4a and c) [2f].

The major elements of broad IL-4/IL-4R α interface carries a charge complementarity which develop polar contacts between crucial residues i.e. Glu9 and Arg88 of A and C helices of IL-4 and Tyr13, Tyr183, Ser70 and Asp72 of IL-4R α [41]. While IL-13 confers a more compact helical scaffold at the interface of IL-13 and IL-4R α (helical turns at A and C helices get shorter). Albeit, the hotspot residues and the binding surfaces of IL-13 are almost identical to IL-4 which are used in association of IL-13 with IL-4R α . Parallel complexation of IL-4 and IL-13 with IL-4R α results in compact overlapping of their four helix bundle. However, IL-13 precisely replace two vital charged residues by exhibiting Glu12 and Arg65 to IL-4R α in corresponding position to establish the same receptor contact as observed in IL-4 residues i.e. Glu9 and Arg88. This residue analogy among IL-4 and IL-13 was investigated by mutagenesis and found to be necessary in their interaction with IL-4R α [43]. Yet, the association of remaining IL-13 contact residues with IL-4R α is distinct according to different binding thermodynamics of both the cytokines with IL-4R α [2f].

3.4. Signaling cascade

IL-4 receptor complexes are deficient of intrinsic kinase activity. Rather, signal transduction is initiated by receptor-associated kinases i.e. a member of JAK family (JAK-1, JAK-2, JAK-3 & Tyk-2). These kinases are linked with each other and some other proteins

i.e. Fes [45] and SYK [46] in the cytoplasmic tail of receptor [44]. This Fes and SYK are protein kinases having essential role in cellular signaling pathways and play key role in activation of IRS-2 in IL-4 signaling pathway. Ligand-stimulated dimerization of receptor complex initiates the phosphorylation of target tyrosine residues in the cytoplasmic tails of both IL-4 receptor chains (IL-4R α and γ -chain/IL-13R α) [28,47] which leads to auto- [48] and *trans*-phosphorylation of JAKs [49]. These tyrosine phosphorylation sites serve as docking territory for subsequent signaling molecules at the cytoplasmic tail of all three IL-4 receptor subunits. The IL-4R α involves five prime tyrosine residues Y1-Y5 that have been well marked in terms of cellular function outcomes and interaction with successive signaling molecules [50]. Y1 is present at the midpoint of an insulin-IL4R-motif enclosed by phosphotyrosine-binding (PTB) domain of insulin receptor substrate (IRS) protein (Dok-1, Dok-2 & Shc). Y2-Y4 are enclosed by SH2 domain comprising protein as in STAT-6 [50d]. Though, Y5 is an integral element of inhibitory tyrosine motif which is bound by SH2 domain (SHP-1 & SHP-2) where SHP-1 depends upon the cell type.

IRS-1 (expressed hematopoietically) and IRS-2 (expressed non-hematopoietically) are the two principal phosphoprotein species stimulated by IL-4 activation depending upon their cellular expression levels and types. IRS-2 is an immense cytoplasmic connector protein which includes PTB, Shc and NPXY linking domain at N-terminus with various potential phosphorylation sites for tyrosine [51], serine and threonine [52]. Probably because of its adjoining to IL-4R coupled JAKs, IRS becomes massively tyrosine phosphorylated after association of IRS-2 PTB subunit with IL-4R motif of IL4-R α receptor chain. Heller et al. illustrated that γ chain of IL-4 receptor is necessary for vigorous tyrosine phosphorylation of IRS-2 pathway, while type-II receptor complex could not trigger IRS-2 pathway effectively [53]. The pY residues of IRS-2 can lead to the activation of various signaling cascades like SOS/Ras/Gap, PI3 K/Akt, and PKB/mammalian by engaging SH-2 subunit along with P85, Grb2 and SHP-2 molecules. This capability of IRS-2 to augment various signaling molecules concedes this IRS-protein to induce mitogenesis, stimulate cell survival [14c], control reproduction, gene expression [54], and insulin responses/energy homeostasis [55]. Generally, in spite of IRS-2 correlation with Grb2, IL-4 and IL-13 do not initiate Ras/MAPK pathway (with few exceptions) [53,56].

IL-4R α associates with IL-4/IL-13 and preeminently initiates tyrosine phosphorylation of STAT-6 which can also stimulate other members i.e. STAT-3, STAT-1, STAT-5, but to minor degree. STAT-6 leads to homo-dimer formation after pY541-SH2 domain

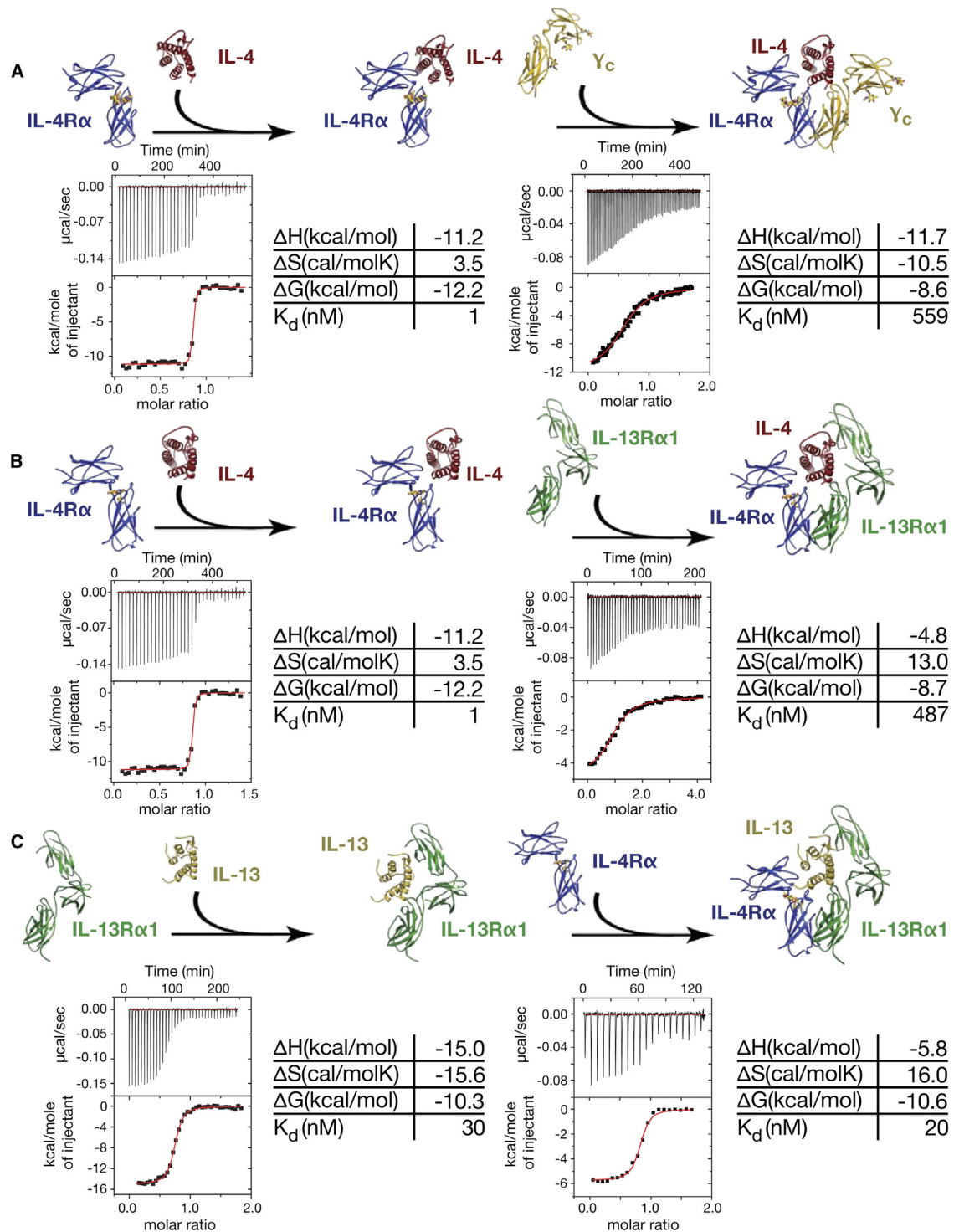


Fig. 4. Comparison of binding thermodynamics, assembly and stabilities of three ternary complexes. Each row represents the reaction order in which the binary and ternary complexes are assembled. (a) Type I complex: reaction diagram for association of IL-4R α and IL-4 followed by recruitment of the second receptor, γ_c . (b) Type II IL-4 complex: the interaction of IL-4R α and IL-4 followed by the recruitment of the second receptor, IL-13R $\alpha 1$. In both (a) and (b), the extremely high affinity of IL-4/IL-4R α ($K_d < 1$ nM) is an estimate due to the steepness of the ITC trace. (c) Type II IL-13 complex: the interaction of IL-13R $\alpha 1$ and IL-13 followed by the recruitment of the second receptor, IL-4R α (LaPorte, S.L.; Juo, Z.S.; Vaclavikova, J.; Colf, L.A.; Qi, X.; Heller, N.M.; Keegan, A.D.; Garcia, K.C., Molecular and structural basis of cytokine receptor pleiotropy in the interleukin-4/13 system. *Cell* **2008**, 132 (2), 259–272).

association [57] and moves this homo-dimer complex towards the nucleus to stimulate transcription by linking to a set of symmetrical recognition elements, TTC-GAA separated by four nucleotides [58]. Usually, STAT-6 forms cooperative association

with other transcription factors and co-activators i.e. NF- κ B or p300/CEBP. Other post-translational modifications of STAT-6 that influence its transcriptional activities involve methylation [59], acetylation [60], and serine phosphorylation [61]. STAT-6 plays a

fundamental role in gene regulation, T-cell survival and proliferation, Th2 differentiation and controlling number of allergic responses by IL-4 & IL-13 [62].

The SH2 domain of protein tyrosine phosphatases, SHP-1 and SHP-2, joined to the presumed ITIM (immuno-receptor tyrosine-based inhibitory) motif enclosing fifth tyrosine (Y713) in stimulated receptor [63]. SHP-2 is extensively expressed while the expression of SHP-1 is hematopoietically confined. SHP-1 tyrosine phosphorylation enhances its phosphatase agility [64], while SHP-2 phosphorylation generates binding site for proteins having SH2 subunits such as IRS-1, p85, Grb2 [65]. Irina G. Luzina et al. verified that SHP-1 reduces STAT-6 signaling by hydrolyzing phosphatidylinositol (3–5) triphosphate and the product of PI3 kinase activity via type-I receptor complex [17,66]. Reduction in IL-4/IL-13 signaling is also achieved by accelerated activation of SOCS (suppressor of cytokine signaling) proteins [67]. SOCS-1 & 3 suppress IL-4 signals by halting JAK activity precisely via their kinase inhibitory domain [68] when bound to IL-4R α [69]. Consequently, it prevents the association of subsequent signaling proteins to the receptor [70] and promotes the dissipation of JAKs and signaling proteins [71].

3.5. Comparison of receptor activation between IL-4 & IL-13

IL-4 and IL-13 interacting with type I & type II receptor complexes causes STAT-6 incitement [2e]. Sherry et al. correlated IL-4 and IL-13 stimulated phosphorylation of STAT-6 to check the receptor complex activation by FACS (Fluorescence Activated Cell Sorting). They evaluated signaling of type-I receptor complex in human B-cell lines (Ramos), displaying γ chain and IL-4R α , while type-II receptor complex (exhibiting IL-4R α and IL-13R α -1-1) were evaluated in human epithelial carcinoma cell lines (A549). Treating carcinoma cell lines i.e. A549 with IL-4 and IL-13 for 30 min, robustly activated the tyrosine phosphorylation of STAT-6. IL-4 promptly induces phosphorylation at markedly lower doses, approximately 5–10 fold, than IL-13. While analogous response was observed by IL-4 in normal B-cell lines i.e. Ramos to activate tyrosine phosphorylation of STAT-6.

IL-4 promptly stimulated STAT-6 tyrosine phosphorylation in A549 attaining plateau levels within 10–15 min of induction. However, the activity induced by IL-13 was extensively mild, requiring more time to establish plateau. Exposing A549 cells to 1 ng/ml, 5 ng/ml & 50 ng/ml of IL-13 (?) induces STAT-6 tyrosine phosphorylation to elevated levels at >60, 30, & 15 min, correspondingly. The comparative delay in STAT-6 phosphorylation by IL-13 through type-II receptor complex was ultimately considerable at minor quantity of cytokine i.e. 1 ng/ml. Maximum phosphorylation was accomplished by IL-4 between 10 and 15 min [2f].

4. Role of IL-4 in disease pathogenesis

IL-4 and its signaling system play significant role in progression of several immune disorders mainly allergic and autoimmune diseases [72]. It has been illustrated in various animal models that allergic asthma is primarily dependent on Th2 cells activation and their cytokines, particularly IL-4 and IL-13 [73]. IL-4 directed Th2 cell differentiation is also a fundamental step during immune response against parasitic infestations [74]. Additionally, role of IL-4 in cancer is highly complicated [75] as it can support tumor progression [76]. Perhaps, because of its anti-apoptotic activities, IL-4 signaling cascade can trigger reluctance to anti-tumor therapy [5b,14b,76,77]. All these evidence distinctly demonstrate the significance of IL-4 in the advancement of various immune-based diseases [32].

a) IL-4 In allergic diseases: Asthma

Allergic diseases are set of disorders that manifest due to innate immune response and Th2-dependent machinery [78]. Some of the pathological condition that represent this group include atopic dermatitis, allergic asthma, and systemic anaphylaxis. IL-4R signaling in non-hematopoietic cells such as smooth muscle and airway epithelium, plays a pivotal role in promoting allergic airway inflammation [79]. Sources of these diseases by STAT-6 dependent system strictly involve both IL-4 and IL-13 activity [6b,6c,73,80]. These two cytokines are involved in several biochemical and cellular proceedings that are responsible for asthma and allergic diseases [81]. They control Th2 cells, neutrophils, eosinophil, mast cells, goblet cell hyperplasia, adhesion molecules, IgE production and airway hyper-responsiveness as described in Table 1. This is explored by the evidence that IL-4 & IL-13 administration to mice triggers asthmatic symptoms such as mucous production, eosinophilia, and airway hyper-responsiveness [82]. According to some studies, it is verified that IL-4 is crucial during allergen challenges because it plays a vital role in controlling Th2 cell differentiation. On the other hand, IL-13 regulates the subsequent processes. The role of these cytokines in asthma is dependent on STAT-6 stimulation over IL-4R α chain. Hence, mice deficient of this transcription factor are exempted from allergen induced asthma [83].

b) IL-4 In solid tumors

The close relation between IL-4 and tumor progression has been investigated through several diseases [84]. In various tested cancer patients, as versatile as breast cancer [85], prostate cancer [86], non-small cell lung carcinoma [87], renal cell carcinoma [88], and other types of tumors [84], enhanced IL-4 production by immune cells such as peripheral blood mononuclear cells (PBMC) and tumor infiltrating lymphocytes (TIL) has been explored by reverse transcription-PCR (RT-PCR), ELISA, and flow cytometry. Apart from enhanced IL-4 expression in cancer patients, the increased amount

Table 1
Effects of IL-4 and IL-13 on various immune cells leading to the pathogenesis of allergic asthma.

Immune Cells	Cytokines	Effects/Function	Pathogenesis
Airway Smooth muscle (ASM) cells	IL-13	Enhancement of ASM proliferation and contractility	Airway remodeling
B-cells	IL-4	Switching of IgM to IgE antibodies	Activation of Mast cells to release histamine, prostaglandins and other cytokines
Eosinophils	IL-4	Proliferation and recruitment of eosinophils to airway passage	Airway inflammation
Epithelial cells	IL-13	Increase iNOS production	Increase mucus production
Fibroblasts	IL-13	Differentiation of fibroblasts to myofibroblasts	Goblet cell hyperplasia & Airway remodeling
Macrophages/Dendritic cells	IL-4 & IL-13	Enhance proliferation and activation of macrophages & dendritic cells	Airway inflammation
Mast cells	IL-4	Activation of mast cells	Release histamine, prostaglandins and other cytokines

of IL-4 receptor has also been revealed in various cancerous cells. This high IL-4R expression was explored by 125 I-IL-4 binding assay or flow cytometry, involving malignant melanoma [89], renal cell carcinoma [90], head & neck cancers [91], and glioblastoma [92], that frequently express 10,000–13,000 IL-4R/cell with modification depending upon the type of tumor. While, normal cells such as B-cells [93], monocytes [94] and endothelial cells [95] displayed much less number of IL-4R, that is, about 50–500 receptors/cell. Furthermore, massive IL-4R expression was also observed on tissue sections such as breast, bladder and prostate cancer [96]. Zhigaung Li et al. investigated a pair of tumor cells with or without IL-4 responsive activity which showed that endogenous IL-4 can primitively motivate tumor cells and enhanced their resistance to apoptosis by advancing the tumor growth. These observations provide a path to control tumor development at the molecular level and would be valuable for developing unique therapeutic approaches [97].

c) IL-4 In central nervous system malignancies

Brain tumor cells like astrocytoma, glioblastoma, meningioma and medulloblastoma over express various growth factor receptors including epidermal growth factor receptor (EGFR) [98], high-affinity forms of fibroblast growth factor receptor-1 (FGFR-1) [99], transferrin receptor (TfR) [100], insulin like growth receptor [101], IL-13 receptors [29c,102] and urokinase receptor [103]. Additionally, many researchers disclosed the expression of erythropoietin receptor, vascular endothelial growth factor receptor-1 (VEGFR-1) [104], and platelet derived growth factor receptor in CNS malignant cells [105]. Recently, Sachin puri et al. worked on meningioma, illustrated that most of the human meningioma displayed IL-4 receptors massively [106], indicating its role in cancer progression. As human meningioma express IL-4R α and IL-13R α -1, but not surface γ chain, it can be concluded that the majority of meningioma display type-II IL-4 receptors instead of type-I. Several studies confirmed that malignant brain tumor cell lines and high grade brain tumor specimens expressed enhance IL-4R with respect to normal brain cell lines, but it is not evident that IL-4 receptors play any role in tumorigenesis of brain tumors [107]. Probably, IL-4 receptors serve as EGF-VIII growth factor receptors that are greatly displayed in approximately 50% of malignant glial tumors and may promote autocrine and paracrine growth of these tumors [98b].

d) IL-4 In B-cell chronic lymphocytic leukemia (B-CLL)

B-cell chronic lymphocytic leukemia (B-CLL) is characterized by gradual aggregation of clonal (kappa or lambda positive) CD5p B lymphocytes, which causes these lymphocytic leukemia cells to be unresponsive to apoptosis [108]. Various studies illustrated that B and T cells and their culture from CLL patients contained profound IL-4 levels [109]. IL-4 is synthesized by either co-stimulatory T-cells or CLL B-cells. T-cells synthesized IL-4 can be significant for in-vitro progression and growth of neoplastic B-cells. Likewise, TNF- α and IL-2 support B-CLL proliferation [110]. Moreover, factors like IL-4, IFN- γ and IFN- α stimulate viability, survival and inhibition of apoptosis in leukemia cells [14a,111]. IL-4R stimulates PI3 kinase by signaling mechanism, and this kinase defends hematopoietic cells against apoptosis [112]. Moreover, IL-4 plays an important role in enduring [14e] and enhancing anti-apoptotic constituents of BCL-2 [113] in hematopoietic cells intervened by PI3 kinase [18]. The binding affinity of IL-4R α towards tumor cells was found to be stronger when compared to normal cells, as examined by Kawakami and his coworkers [114].

e) IL-4 In rhabdomyosarcoma (RMS)

Rhabdomyosarcoma (RMS) is histologically recognizable soft tissue sarcoma with myogenic phenotype which constitutes local invasiveness, rapid growth with enhanced susceptibility to metastasize. Histologically, RMS has two major types i.e. Embryonic RMS (ERMS) and Alveolar RMS (ARMS) [115]. IL-4 receptor

signaling plays various roles in normal body physiology and disease pathology as well. The cytokine, IL-4, serves as a myoblast recruitment factor to model mature myotubes [116]. In murine breast cancer models, IL-4 and IL-13 synthesized by Th2 cells trigger tumor associated macrophages to promote pulmonary metastasis facilitated by epidermal growth factor secretion [117]. Nanni et al. described the treatment of ERMS cell lines with IL-4 debilitates differentiation ability of these cells and up regulates the migration ability and cell growth [118]. They demonstrated that IL-4 activates ERMS cell proliferation by JAK/STAT signaling cascade. However, in muscle cell differentiation, IL-4 receptor signaling mechanism is necessary for myotubes maturation. Nascent (primary) myotubes made by myoblast–myoblast union make IL-4 (NFAT dependent regulation) to incorporate encompassing IL-4R α myoblast by initiating (secondary) mature myotube generation [116]. Hosoyama. T et al. demonstrated that activation of IL-4 receptor signaling mechanism up-regulates tumor cell proliferation in both ERMS & ARMS. Comparatively, tumor cell proliferation reduces after cessation of IL-4 signaling machinery, but does not promote apoptosis [119].

f) IL-4 In HIV Infection

IL-4 can also play a role in HIV infection and disease advancement. It initiates HIV co-receptor expression i.e. CXCR-4 and DC-SIGN [120], and stimulates HIV execution which can hasten disease pathogenesis [121]. T-lymphocytes with helper activity are the key element of HIV-1 infection. More signs of immune dysregulation like polyclonal B-cell initiation, prior failure of cellular responses to recall antigen and hyper-gammaglobulinemia are found in all HIV-1 infected patients [120b,122] and are associated with Th2 synthesized cytokines (IL-4, IL-5, IL-13). A shift to Th2 response has been assumed to be significant in AIDS pathogenesis [123]. Along with this, Th0 state or atypical system of cytokine production have also been revealed [123,124]. Enhanced production of IL-4 has been detected in T-cell clones of Th2 phenotype from HIV-1 infected patients [125] by intracellular staining [126] and in-situ hybridization [127]. Moreover, a study explored that virus replication increases with high level of IL-4 in lymph nodes from infected children [128].

5. Classification of IL-4 inhibitors

a) Soluble IL-4R α domain

The extracellular domain of cytokine-receptor are synthesized biologically by shedding or by distinct mRNA via diverse splicing [129]. It has been intimated that these soluble receptor mutants can play a significant role in maintaining cytokine activity. This soluble receptor-derived linking protein serves as the physiological proteins that should not evoke immune response. A soluble mutant of IL-4R α interacts with IL-4 with lower affinity of about 2–4 fold than the heterodimer receptor. Hence, soluble binding protein is capable of competing with the cell-bound receptors efficiently for its ligand and is immensely effective in inhibiting IL-4 functions [130]. But, still this soluble IL-4R α is unable to bind with IL-13 to a detectable extent and under some conditions, may promote IL-4 effects through the similar mechanism as detected for anti-IL4 antibodies [131]. Some soluble mutants for IL-4R α are subjected to clinical trials for utilization in allergic asthma therapy [132].

b) Antibodies against IL-4 & IL-4R α

Protein-protein association is inhibited by distinct antibodies fastened at or near the interaction surfaces, by which they prevent the direct contact of natural binding participants. Antibody can be developed against receptor or ligand. Prohibitory antibodies against IL-4 [133] and IL-4R α [134] have been accomplished and were effective in cellular assays and in mouse models of allergy and

parasitic infections. It should be noticed that the inhibitory antibodies against IL-4R α is the only molecule that prevents binding of both IL-4 and IL-13 proteins.

Swart et al., investigated the effects of AMG317 (fully IgG2 monoclonal antibody) on lung inflammation and AHR (airway hyper responsiveness) in murine models of cockroach allergen-triggered asthma to block IL-4R α . It causes reduction in inflammatory mediators and inflammatory cell counts in bronchial alveolar lavage (BAL) and lung tissues, while serum IgE were also restored back to normal [133a,135].

AIR645, a 2-O-methoxyethyl second generation anti-sense drug has been developed that targets mRNA coding IL-4R α and acts as a dual inhibitor of both IL-4 & IL-13. It reduces IL-4 protein at mRNA levels and down regulates the cytokine production in lung inflammation and AHR [136].

Surprisingly, anti-IL-4 antibodies enhance the biological activity of IL-4 by increasing its serum half-life [2b]. These inhibitory antibodies prevent IL-4 from interacting with receptor certainly; however, this prevention is reversible. So cytokine and antibody may detach again with kinetics relying on the off-rate for distinct cytokine-antibody pair. Antibody association enhances the serum half-life of IL-4 by reducing its renal clearance and defends cytokine from proteolytic assimilation [2b,25a,25c].

c) IL-4 As a targeted immunotoxin

Targeted immunotoxins are recent advancement in therapeutic approaches that have developed for the treatment of cancer. In this, cell surface receptor or tumor antigen is directed by chimeric union of protein constitutes of an antibody/ligand with a toxin or lethal gene to kill tumor cells [107]. As IL-4R is expressed massively on tumor cells, a receptor-directed tumor curative agent was developed, with constituents of proposed ligand i.e. IL-4 and modified bacterial toxin from *Pseudomonas* exotoxin (PE) [137]. This PE is a 66 kDa protein composed of three domains: Domain-I interacts with PE receptor, Domain-II activates translocation of toxin to cytosol, while domain-III prevents protein synthesis by suppressing ADP ribosylation of elongation factor-2 [138]. This chimeric toxin (IL-4PE) was synthesized by integration of 5' end of full length PE cDNA having mutated domain-Ia with human IL-4 cDNA which only interacts with cells displaying IL-4R [92].

To authenticate cpIL-4PE as anti-tumor agent, its anti-cancer activity was inspected in animal models of human glioma. A subcutaneous (SC) tumor model testing human glioblastoma cells (U251) in athymic nude mice was made [139]. Based on these preclinical studies, Phase-I clinical trial was commenced to check the tolerability and safety of cpIL-4PE in human malignant glioblastoma (GBM) [139]. Moreover, Phase-II studies are required to test the activity of this agent as a GBM therapy [107].

d) IL-4 Analogues with site-specific mutation

Cytokine antagonist was developed by modifying binding site with the objective to design a variant which can still interact with IL-4 receptor subunit but not with the second receptor the γ -chain/IL-13R α subunit. Antagonistic mutants are extensively specific and cannot stimulate receptor. The IL-4 receptor machinery is exemplary for modifying protein-protein association via site specific alteration. IL-4 is not only of tremendous significance in case of model system for cytokine-receptor structure [2f,36,140] and signaling [1,141], but also of great therapeutic importance as it is one of the primary regulatory cytokine which mediates growth and cell differentiation [142]. On these terms, mutant IL-4 antagonists were developed by manipulating few residues in the binding site of γ chain subunit [143]. Primarily, three residues R121, Y124 & S125 positioned near the C-terminus were markedly subjected to mutation [35,144]. Meanwhile, modification of R121 and Y124 to aspartic acid i.e. R121D & Y124D developed a protein that

exhibited great binding affinity for IL-4R α with negligible biological activity and resulted in cessation of both IL-4 and IL-13 signaling cascade completely [133a].

The above described IL-4 double mutant (IL-4DM) is also known as Pitracinra which halted the dimerization of surface γ chain or IL-13R α -1-1 with IL-4R α . This IL-4DM modify the endogenous IL-4 (wild type) for interacting with receptor. It is synthesized as receptor antagonist and beneficial for treating eczema and eosinophilic asthma [144,145]. Viswandham with his colleagues developed IL-4DM by chemical mutation of hot spot residues which is highly potent for decreasing immunogenicity, and served an additional benefit of remarkable pharmacokinetics in-vivo [146].

e) Small molecular inhibitors

As proteins have supremacy of evolution-selected specificity that causes little applicability of pharmaceutical agents. Massive synthesis of recombinant proteins is very costly and laborious process. Similarly, delicately constructed agents may rapidly become degraded by proteases and oral application is yet useless. Small synthetic molecules that provide enhanced binding affinity and imitate protein binding surface would be an adorable substitute [2b]. Up till now, few natural but almost no synthetic molecules have been established against IL-4/IL-4R α /IL-13 system.

Few natural molecules acquiring anti-allergic activities, i.e., against IL-4/IL-4R α /IL-13 system constitutes a group of flavonoids which are universally present in fruits, tea and vegetables. Flavonoids suppress IL-4 and IL-13 synthesis, histamine release and basophilic CD40 ligand expression. Investigation of 45 flavones in year 2007, revealed their possibility as anti-IL-4 lead agents. Flavonols and relevant compounds such as ayanin, luteolin, fisetin and apigenin were found to be active inhibitors for IL-4 with IC₅₀ value of 2.5 μ M and exhibited basic structure for preventive activity. This suppressive activity of flavonoid is probably because of stimulation of activated T-cells nuclear factors and AP-1 by down regulating mRNA expression of IL-4, IL-13 and CD40 ligand [147].

6. Future perspective of IL-4/IL-4R α inhibitors

Up till now, various IL-4 variants have been designed which are in phase-II clinical trials. This rapid development of IL-4 antagonists put forward a good example of structural and functional importance of IL-4/IL-4R α complex with its significance in molecular medicine. Detailed information about the structural and binding chemistries of IL-4 and its receptor IL-4R α , enable us to design various mini-proteins that resemble IL-4 protein. These synthetic peptides bind to IL-4R α with K_d = 5 μ M which is very low affinity to use these peptides therapeutically. Apart of it, because of large size and molecular weight of peptides, it exhibited various pharmacological problems such as oral applicability difficulties and evoked immune responses.

Considering all above information, there is a strong need for developing inhibitory analogs that are completely different from wild type IL-4 protein. These small IL-4 analogs capable of binding with high affinity to IL-4R α and altercate with IL-4 and IL-13 would be an exemplary drug. Our ongoing computational studies on IL-4R α reveals that IL-4R α exhibits three binding clusters. Separate ligand for each binding cluster is being designed. Linking these three ligands in a molecule might lead to the development of a high affinity ligand with good pharmacological properties.

Conflict of interest

The authors have declared that no competing of interest exists.

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